

How to Use the Data

1. Load the Data

You can load each dataset into your preferred analysis environment (Python, R, etc.) using pandas or similar libraries. Example in Python:

```
python

import pandas as pd

logs = pd.read_csv('router_logs_15_days.csv')
topology = pd.read_csv('network_topology.csv')
config_changes = pd.read_csv('configuration_history.csv')
app_usage = pd.read_csv('application_usage.csv')
user_activity = pd.read_csv('user_activity.csv')
external_events = pd.read_csv('external_factors.csv')
```

2. Explore and Analyze

- Use the router logs to identify traffic patterns, congestion points, and anomalies.
- Leverage topology data to understand network structure.
- Incorporate configuration changes and external events to contextualize network behavior.
- Analyze application and user activity to predict peak loads.

3. Build Predictive Models

- Use historical data to train models that predict congestion or outages.
- Develop algorithms for dynamic bandwidth reallocation based on predictions.

4. Visualization

- Create dashboards or visualizations to monitor network health.
- Plot traffic trends, latency, and congestion events over time.

Tips for Participants

- Focus on feature engineering from the datasets to improve model accuracy.
- Experiment with different machine learning algorithms (classification, time-series forecasting).

- Validate your models with unseen data.
 - Document your approach, assumptions, and findings.
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Additional Notes

- The datasets are synthetic but designed to resemble real-world network data.
- Feel free to extend or customize the datasets for more complex scenarios.
- Use the provided documentation (datasets_description.md) for detailed dataset descriptions.